6.7.6 Scattering of sunlight

When sunlight enters the atmosphere of earth, the particles present in the atmosphere change the direction of the light. This process is known as scattering of light.

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If the scattering of light is by atoms and molecules which have size *a* very much less than that of the wave length λ of light, ($a<<\lambda$), then the scattering is called *Rayleigh's scattering*. The intensity of light scattered in Rayleigh's scattering is inversely proportional to fourth power of wavelength.

$$I \propto \frac{1}{\lambda^4} \qquad (6.98)$$

According to equation 6.89, during day time, violet colour which has the shortest wavelength gets more scattered then the other colours. The next scattered colour is blue. As our eyes are more sensitive to blue colour than violet colour, the sky appears blue during day time as shown in Figure 6.45(a). But, during sunrise and sunset, the light from sun travels a greater distance through the atmosphere. Hence, the blue light which has shorter wavelength is scattered away and the red light which has longer wavelength and less-scattered manages to reach our eye. This is the reason for the reddish appearance of sky during sunrise and sunset as shown in Figure 6.45(b).

If light is scattered by large particles like dust and water droplets present in the atmosphere which have size *a* greater than the wavelength λ of light, ($a >> \lambda$), the intensity of scattering is equal for all the colours. This non-Rayleigh's scattering is independent of wavelength. It happens in clouds which contains large amount

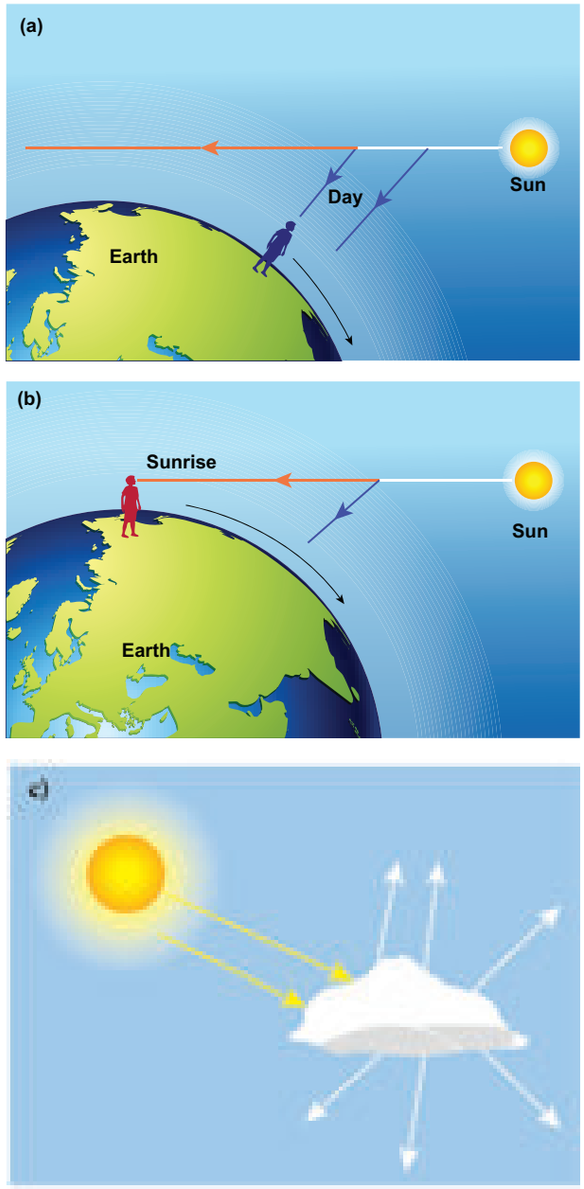


Figure 6.45 Scattering of different types

of dust and water droplets. Thus, in clouds all the colours get equally scattered. This is the reason for the whitish appearance of cloud as shown in Figure 6.45(c). But, the rain clouds appear dark because of the condensation of water droplets on dust particles that makes the cloud opaque.

If earth has no atmosphere there would not have been any scattering and the sky would appear dark. That is why sky appears dark for the astronauts who could see the sky from above the atmosphere.

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